Geophys. J. Int. (1989) **99**, 521–531

New results and a palaeomagnetic overview of the Pontides in northern Turkey

Mustafa Sarıbudak
İ.T.Ü. Maden Fakültesi Jeofizik Müh. Bölümü, Ayazağa Kampüsü, Ayazağa 80626, İstanbul, Turkey

Accepted 1989 May 5. Received 1989 May 5; in original form 1988 December 16

SUMMARY

Palaeomagnetic results are reported from 16 sites within volcanic and volcano-sedimentary rocks of Cretaceous and Eocene age from the central Pontides in northern Turkey. The sampled area is bounded by the North Anatolian Fault (NAF) to the south and the Black Sea to the north. The palaeomagnetic data collected demonstrate the presence of both non-rotated and anticlockwise rotated regions; (i) declination directions including Lower–Upper Cretaceous and Eocene sites do not indicate any significant rotations north of where the NAF exhibits an arcuate shape, (ii) 20° anticlockwise rotation was observed in the Sinop peninsula, the northernmost extremity of the central Pontides, (iii) in the west of the central Pontides, two sites with normal and reversed antipodal directions indicate a 20° anticlockwise rotation and (iv) another region lies to the east of the study area and shows a rotation in an anticlockwise sense of about 5°–30°. The Upper Cretaceous and Eocene directions reported in this paper and other palaeomagnetic studies show that there is not much difference between the inclination values of the Upper Cretaceous ($I = 40.1°$, $\alpha_{95} = 7.2°$) and the Eocene sites ($I = 39.6°$, $\alpha_{95} = 8.1°$). Furthermore, the values of the inclination available from both sedimentary and volcanic Upper Jurassic–Lower Cretaceous formations are shallower than the Upper Cretaceous ones. This suggests that the entire Pontides have undergone a northward drift of over 2000 km since the Upper Jurassic.

Key words: geological evolution, palaeomagnetism, Pontides, Turkey.

1 INTRODUCTION

Few palaeomagnetic data from Turkey have been published so far and those that have are mostly from the Pontides, the northern-most tectonic unit in Ketin's (1966) scheme. Şengör & Yılmaz (1981) later distinguished, within what Ketin called the Pontides, a Rhodope–Pontide fragment and a Sakarya continent.

The first study carried out in the western Pontides was on the Amasra redbeds of Permo-Triassic age by Gregor & Zijderveld (1964). These Permo-Triassic rocks gave a magnetization direction of $D = 292°$ and $I = -14.8°$, which was significantly different from expected African or European directions. Sarıbudak, Sanver & Ponat (1989) reported preliminary results on Lower Triassic volcanic and sedimentary rocks of the western Pontides indicating that the western Pontides formed in Laurasia rather than Gondwana during the Triassic.

A number of studies were undertaken on the Mesozoic and early Tertiary rocks of the eastern Pontides (Van der Voo 1968; Orbay 1978; Orbay & Bayburdi 1979; Baydemir 1982). All results obtained from these studies were interpreted assuming that Turkey (not only the Pontides) moved as a single unit together with the Arabian promontory and Africa since late Mesozoic time.

Lauer (1981) reported a set of palaeomagnetic data from Turkey some of which were obtained from the western Pontides. In interpreting his data, Lauer divided Turkey into three blocks: (i) Pontides, (ii) western Taurides and (iii) eastern Taurides, each of which was independent through geological time and became one single unit by convergence of the African and Eurasian plates in the Neogene. Even though the constraints on the palaeolatitude of the Pontides throughout geological time have been poor owing to insufficient data, Westphal *et al.* (1986) pointed out that palaeolatitudes of the Pontides have a closer affinity with the African palaeolatitude curve than the Eurasian curve, at least since the late Jurassic.

Geologically, several contrasting views exist on the pre-Jurassic geodynamical evolution of the Pontides (Şengör, Yılmaz & Sungurlu 1985; Robertson & Dixon 1985; Adamia *et al.* 1980; Bergougnan & Fourquin 1980; Üşümezsoy 1987). For post-Jurassic time it was pointed out that the Pontides (excluding the Sakarya continent)

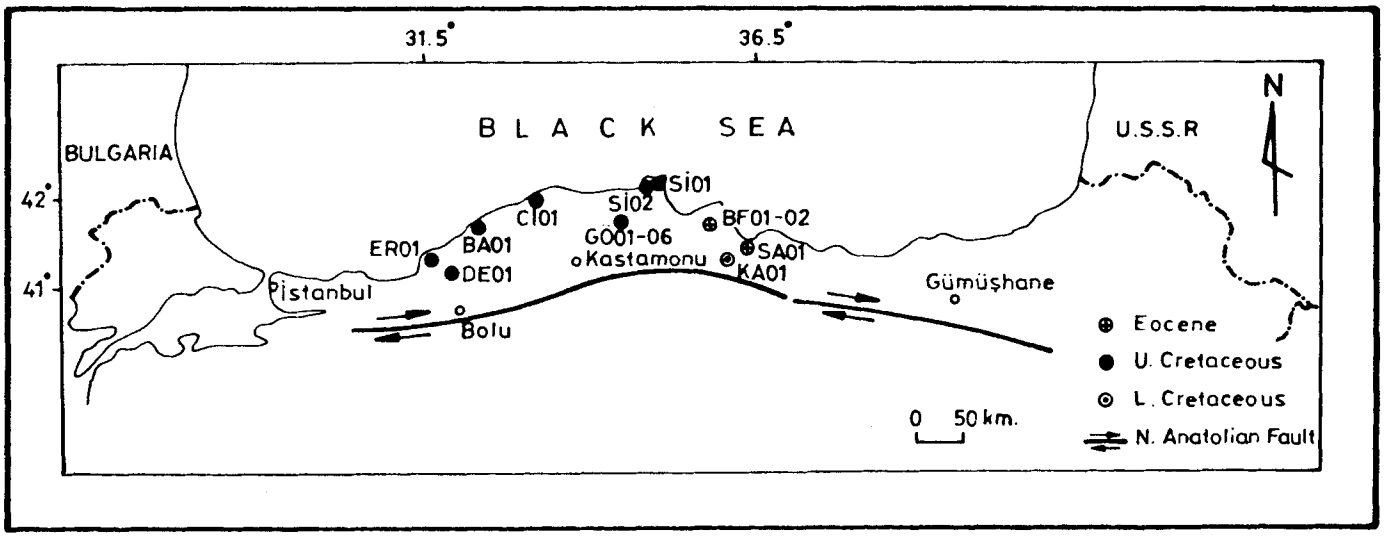


Figure 1. Location of the sites in the Pontides.

constituted a south-facing passive continental margin which was converted into an active margin in Aptian time along with the development of a south-facing magmatic arc until the Eocene to the north of Neo-Tethys and the Pontides were assumed to be the part of Eurasia during Upper Jurassic onwards (Şengör & Yılmaz 1981; Şengör *et al.* 1985; Görür 1988).

In this paper, results from Lower–Upper Cretaceous and Eocene volcanic rocks of the Pontides are reported. The purpose of this study is both to provide new data and review the available palaeomagnetic directions and interpret them in terms of possible tectonic translations and rotations.

2 GEOLOGY OF THE SAMPLED SITES

The location of the sampled sites is shown in Fig. 1. A total of 95 oriented blocks were collected from the central Pontides whose ages range from the Lower Cretaceous to the Eocene. A description of the sampled sites and the age criteria used for each site are given in Table 1.

3 LABORATORY TECHNIQUES

Palaeomagnetic measurements were made at the Kantek laboratory, which is run by the İTÜ and the Kandilli Observatory. In the laboratory, cores with a diameter of 25 mm were drilled from the blocks, and all cores were cut into specimens 22 mm long. Three or four specimens were prepared from each blocks, two of which were selected as pilots to be treated by alternating magnetic field (AF) and thermal demagnetization techniques in order to isolate the stable magnetization from the secondary magnetizations. Specimens treated by progressive AF demagnetization were demagnetized in seven to nine successive steps up to 50 mT. Specimens selected for thermal treatment were heated up to 550 °C in 10–11 steps. After each demagnetization step, the magnetization of the specimens was measured on an astatic magnetometer. Thermal demagnetization was applied to the remaining specimens in fewer steps.

Analysis of the demagnetization behaviour of the samples was undertaken with the help of vector diagrams, stereographic projections and normalized intensity curves. Specimens which showed undirectional decay of the magnetization on higher levels of demagnetization were considered stable. On this basis, a characteristic direction was determined for each specimen. These directions were combined using Fisher (1953) statistics to calculate site-mean directions.

4 PALAEOMAGNETIC RESULTS

By using progressive demagnetization on the specimens, good quality demagnetization diagrams were obtained. Fig. 2 shows typical results. In all cases, the direction obtained by thermal demagnetization was indistinguishable from that obtained from AF demagnetization. These cleaning tests revealed that all specimens were characterized by a small low coercivity or low blocking temperature secondary component that was easily removed either by AF demagnetization of 15–25 mT or by thermal demagnetization of 200–250 °C. The decay of the remanence intensity during progressive demagnetization is also illustrated in Fig. 2.

The results obtained are summarized in Table 2. Stereographic projections from each site (a) NRM, (b)

Table 1. Geological description of the sampled sites from the central Pontides.

Site	Formation name	Rock type	Number of blocks	Bedding direction	Age	References
KA01	Çağlayan	Andesite	6	302.51 SE	Aptian	Ketin & Gümüş (1963)
GÖ01–06	Gökçeağaç	Andesitic tuff	25	274.25 NE	Upper Cretaceous	Yilmaz & Tüysüz (1984)
Cİ01	Yemişli	Basalt	10	270.47 N	Upper Cretaceous	Ketin & Gümüş (1963)
BA01	Cambu	Andesite	8	40.11 SE	Campanian	Şentürk & Özçelik (1983)
ER01	Cambu	Andesite	8	60.14 SE	Campanian	
DE01	Cambu	Andesite	10	225.26 NW	Campanian	
Sİ01–02	Hamsoros	Basalt	14	173.23 NE	Upper Cretaceous	Ketin & Gümüş (1963)
BF01–02	Kusuri	Basalt	10	265.26 NW	Eocene	
SA01	Kusuri	Basalt	4	295.21 NE	Eocene	

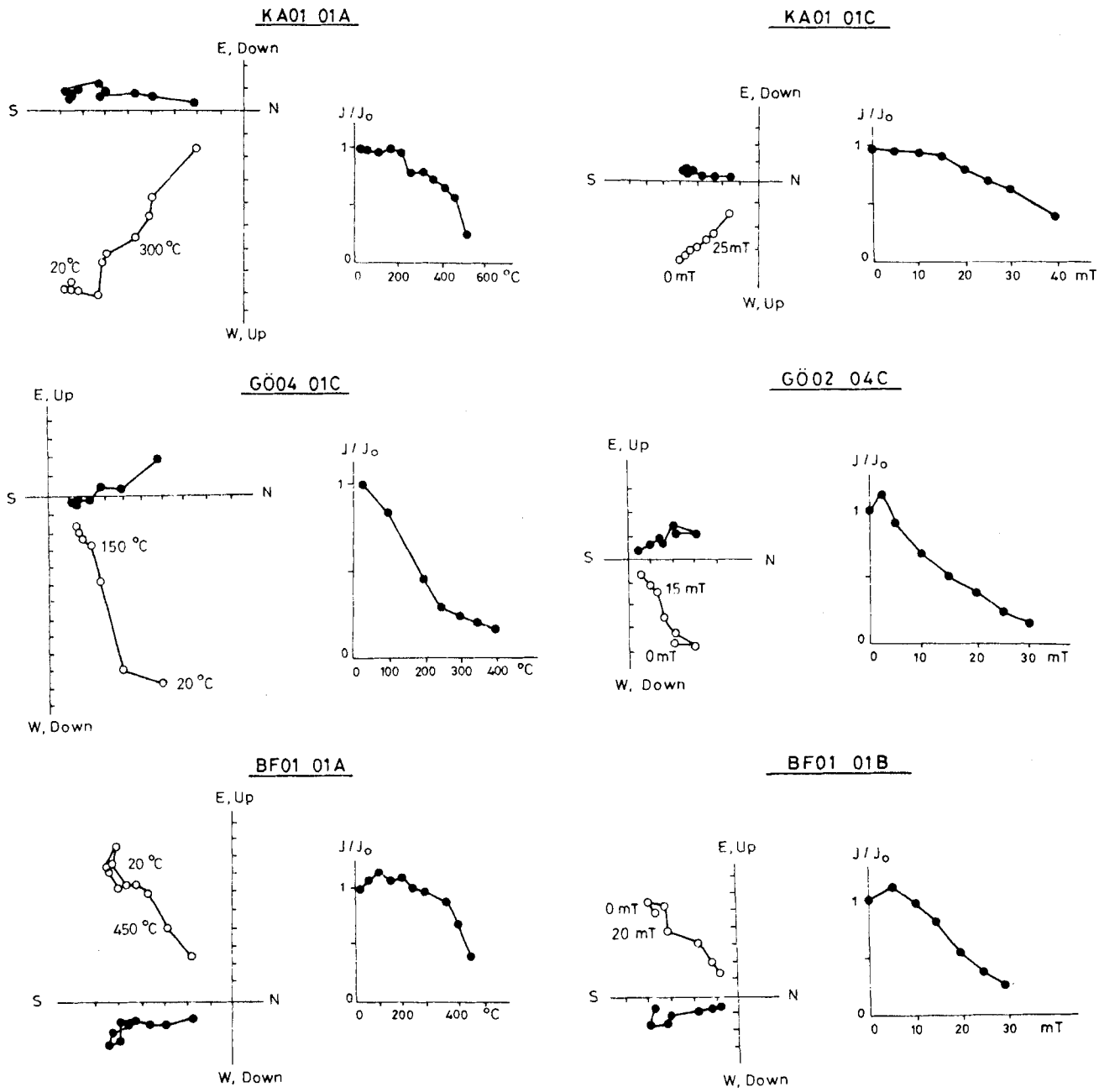


Figure 2. Typical thermal and alternating fields demagnetization diagrams. Full circles: horizontal projections; open circle: vertical projection. Jo: NRM intensity. The unit on the vector plot scales is 0.2 A M^{-1}.

Table 2. Palaeomagnetic results from the central Pontides. n/N: number of samples accepted/taken.

		Before bedding correction		After bedding correction			
Site	n/N	D (°)	I (°)	D (°)	I (°)	k	α_{95} (°)
KA01	5/6	177.4	−420.0	186.3	3.2	190.0	5.6
GÖ01–06	23/25	1.6	54.4	1.9	29.4	243.0	4.3
CI01	6/10	207.4	−72.6	188.3	−27.0	69.0	8.1
BA01	6/8	173.1	−45.3	182.8	−53.4	67.1	8.2
ER01	5/8	160.0	−40.0	163.3	−56.7	20.0	17.5
DE01	5/10	356.4	59.0	339.3	37.0	63.3	9.7
SI01	6/8	176.0	−44.3	154.9	−39.0	157.2	5.3
SI02	6/6	177.6	−35.7	161.3	−33.0	44.0	10.2
BF01	5/5	186.9	−53.6	182.0	−28.0	194.0	5.5
BF02	5/5	173.0	−58.3	172.7	−32.3	243.6	4.9
SA01	2/4	123.0	−46.0	141.1	−39.4	—	—

(primary directions), (c) tilt corrected RM directions of blocks, are shown in Figs 3 and 4. These illustrate the changes in the magnetization direction as a result of cleaning and bedding correction.

Figure 3 shows that, after bedding correction, the palaeomagnetic directions obtained from the rocks of Lower Cretaceous, Upper Cretaceous and Eocene age are N–S indicating a null rotation of the region. The rotated sites are shown in Fig. 4. More scattered sites Si01–02 which are located in the Sinop peninsula represent an approximately 20° anticlockwise rotation, as do sites ER01 and DE01 from east of the central Pontides, which also have antipodal

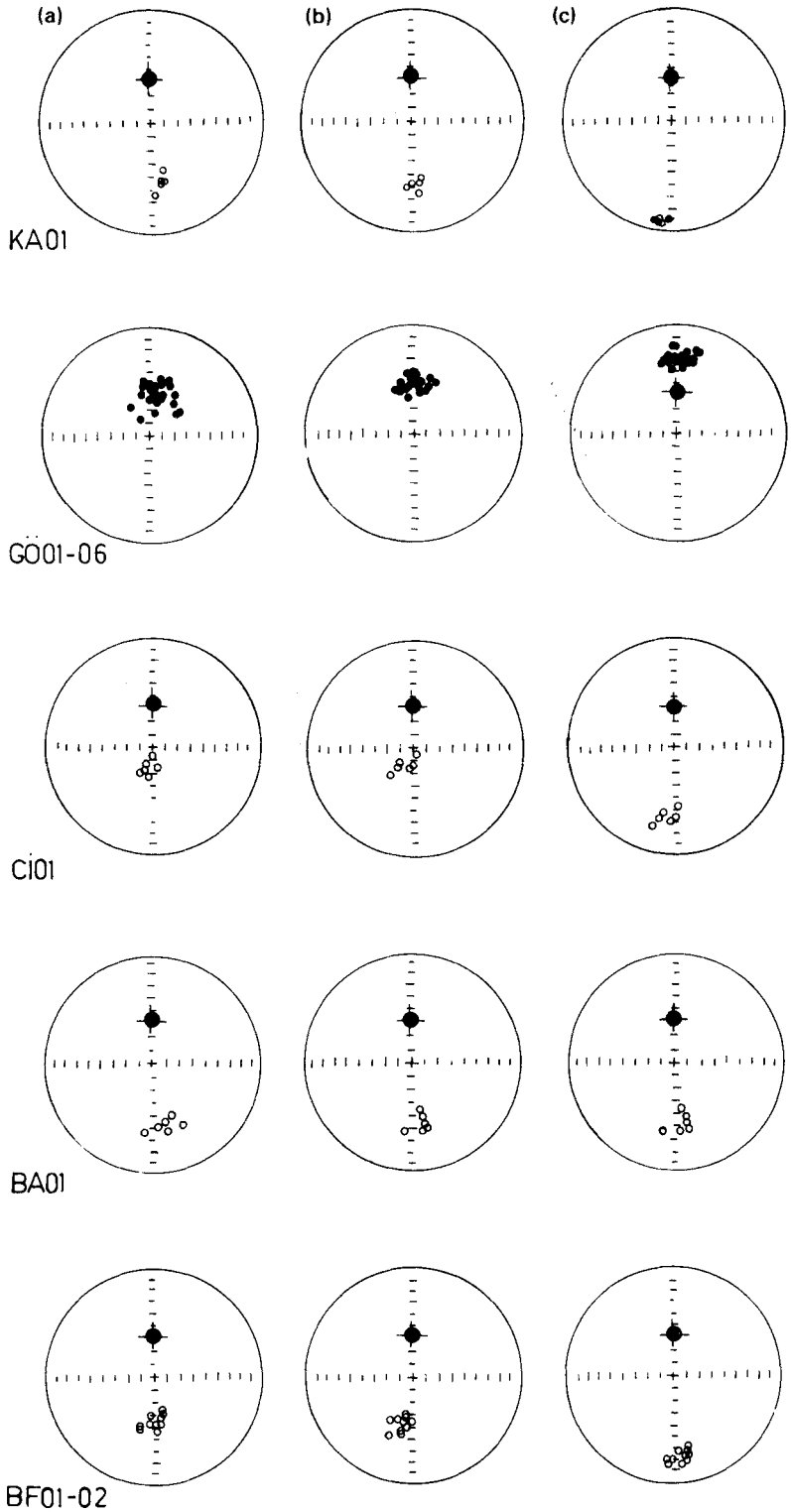


Figure 3. Stereographic projections of the directions of magnetization from the non-rotated sites. (a) NRM; (b) after demagnetization; (c) after demagnetization and tectonic correction. Full circles: normal directions; open circle; reverse direction; + denotes the present-day magnetization direction.

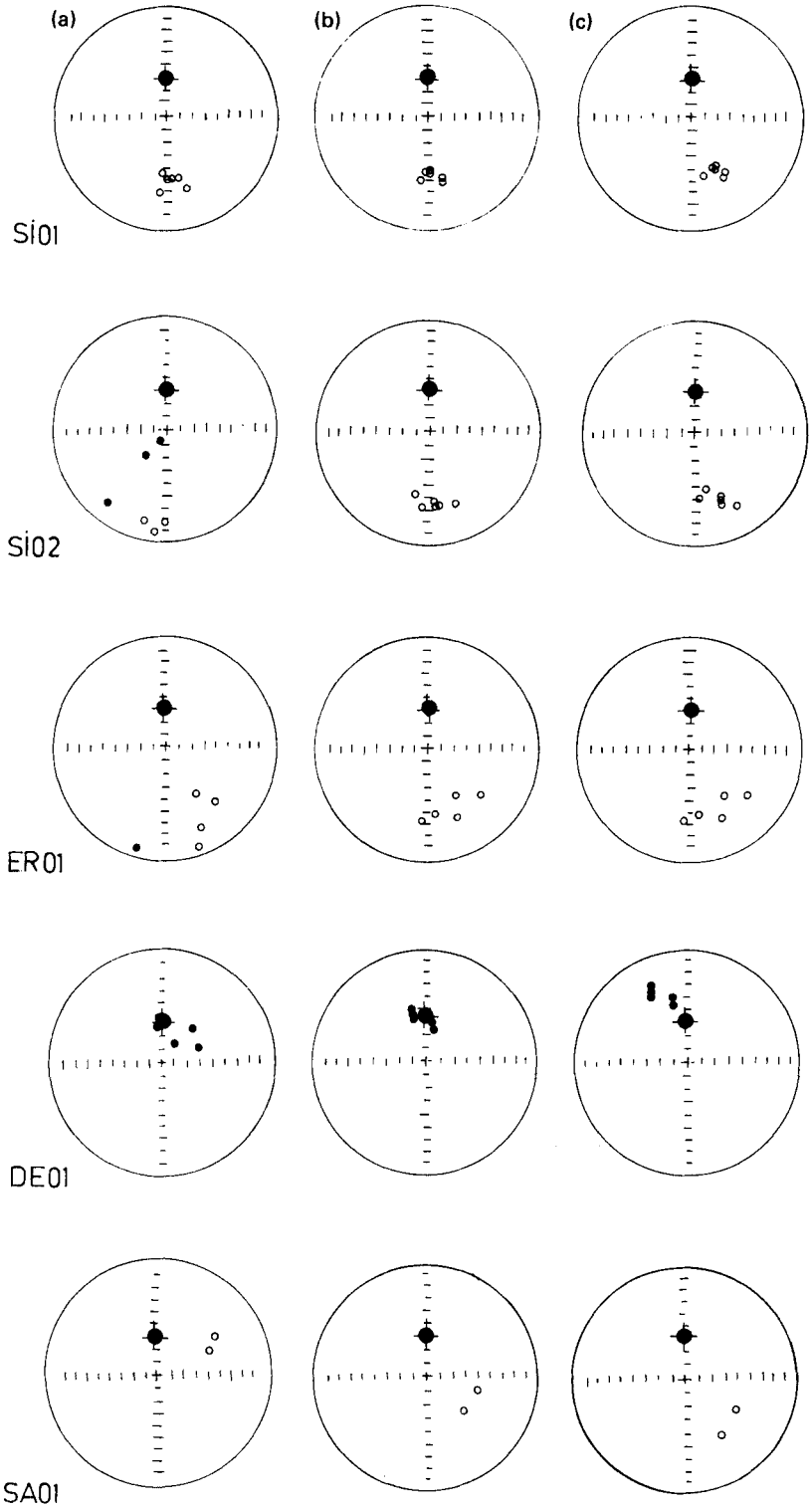


Figure 4. Stereographic projections of the directions of magnetizations from the rotated sites. For explanation, see Fig. 3.

directions. Site SA01, from which two blocks were taken also indicates an anticlockwise rotation but it is of low reliability; it is reported here because the rotation pattern is consistent with previous results obtained adjacent to the study area.

5 INTERPRETATION AND DISCUSSION

Thus all the published and some unpublished palaeomagnetic data, which were obtained through the joint work of Centre des Faibles Radioactivités (CFR) in Gif sur Yvette,

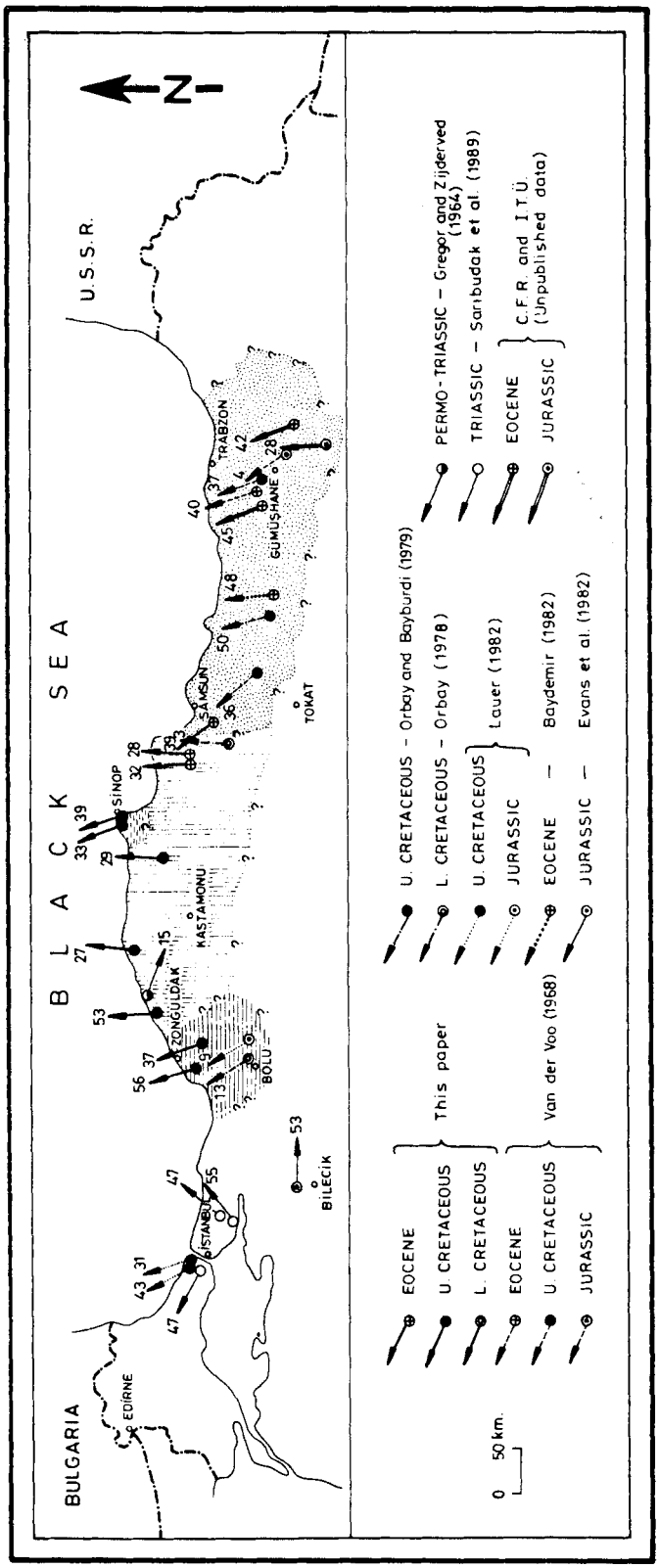


Figure 5. Observed Jurassic, Cretaceous and Eocene palaeomagnetic directions at sites in the Pontides. The reversed directions are inverted through the origin.

France and İTÜ, are shown in Fig. 5 for comparison with the results reported in this study.

By examining Fig. 5, the Pontides can be divided into several distinct regions on the basis of declination directions. (1) The first region lies between Bartın (BA01) and Samsun (SA01), which corresponds to what is called the central Pontides, in which all sites whose ages range from Lower Cretaceous to the Eocene do not show any significant rotation. However, the Permo-Triassic declination shows a clockwise rotation of 110° in the Amasra region, suggesting that the region had experienced a large clockwise rotation between the Permo-Triassic and the Upper Cretaceous. (2) Two Upper Cretaceous sites (Sİ01–02) from the Sinop Peninsula has undergone a 20° anticlockwise rotation. (3) Several sites sampled in Jurassic, Cretaceous, and Eocene rocks distributed within the eastern Pontides show that the eastern Pontides have undergone an anticlockwise rotation of between 5 and 30° (Van der Voo 1968; Orbay & Bayburdi 1979; Baydemir 1982; this paper, and unpublished report of CFR & İTÜ). (4) West of the central Pontides covering the region of Bolu and Ereğli (ER01), four sites (Orbay 1978; Lauer 1981; and this paper) give very comparable directions indicating an anticlockwise rotation of 20°. (5) A clockwise rotation of about 40° since the Triassic was reported from the southeast of İstanbul, and one single direction from Triassic rocks in northwest of İstanbul indicate an anticlockwise rotation of 68° (Sarıbudak *et al.* 1989). Two Upper Cretaceous directions obtained near the Triassic site also give a 20° anticlockwise rotation on average (Lauer 1981). (6) From the Upper Jurassic Bilecik limestones in southwestern Pontides Evans *et al.* (1982) documented a 90° clockwise rotation of the area since the late Jurassic (see Fig. 5).

To constrain the timing of the rotations observed in the Pontides it is necessary to sample more units of younger ages. However, a correlation between of declinations of all available data and the neotectonic features in the Pontides is instructive (Fig. 6). (a) In the eastern Pontides the axes of the folds and thrusts strike NE–SW and north western declinations are characteristic. The causes of the anticlockwise rotation indicated by the declinations may be explained by the post-collision following the indention of the Arabian platform with the eastern Turkey during the Miocene time. The inferred limit of this rotated region toward the west is approximately delineated by the palaeomagnetic results and by the volcanic belt of the Pontides. (b) The central Pontides are characterized by E–W trends of folds and faults which are found compatible with the non-rotated declinations. The western limit of the non-rotated region is not clear due to an insufficient number of sites and the absence of structural observations. (c) However, in the north of central domain, the anticlockwise rotation of the Sinop peninsula can be explained by the normal and thrust faults which bound the region from south and north, respectively (İlhan 1976). (d) In the west of the central Pontides the fold axes also strike NE–SW in agreement with the anticlockwise rotation of the region suggested by the declination data. It is difficult to extend this region toward the west because the two sites from the Gebze region indicate a 40° clockwise rotation which may be related to the NAF initiated during the late Miocene.

The mean inclination values (calculated from Fisher statistics) of rocks of the same age can be used to estimate changes of palaeolatitude of the Pontides from Upper Cretaceous to Eocene. There is no significant difference observed in the averaged inclination values for the Upper Cretaceous and the Eocene which are 40.1° and 39.6°, respectively (Table 3), which position the Pontides at about 23° N palaeolatitude.

The Jurassic and Lower Cretaceous inclination data from the Pontides are difficult to interpret (Table 3). However, the inclination can still be analysed to extract some palaeolatitude evidence. The inclination value of the Bilecik limestone ($I = 53°$) is higher than other inclination data. Accordingly it is debatable whether the Sakarya continent and the Pontides have always been part of a single tectonic unit or whether they have moved independently during most of their history (the first Turkish geotraverse project, organization and program, 1987, TÜBİTAK, pp. 14). However, this high inclination value of the Upper Jurassic limestone could be explained if the Sakarya continent was not attached to the Pontides during the late Jurassic. So that the Bilecik limestone should be excluded from consideration with the Pontides. The Lower Jurassic result (Van der Voo 1968) was also not considered reliable by the author himself. More data have been collected (Orbay 1978; Lauer 1981; present paper; and the unpublished report of CFR & İTÜ). Although these inclination values are not well constrained (Table 4), they strongly suggest that the Pontides were positioned at a lower latitude than the Upper Cretaceous–Eocene times.

As stated before, the Permo-Triassic and Lower Triassic data place the western Pontides at the southern margin of Laurasia. An excursion to the south of the western Pontides seem to have taken place during the Triassic–Jurassic period (Fig. 7). From the geological point of view, Pre-Upper Jurassic period was a critical turning point in the development of the Pontides: The so-called Palaeo-Tethys closed during the early Jurassic. Its oceanic remanents were scattered at the Küre–Kastamonu region in the central Pontides (Şengör & Yılmaz 1981; Aydın, Serdar & Şahintürk 1982). According to Şengör, Yılmaz & Ketin (1980) the continental assemblage of the İstanbul Nappe (western Pontides) rested on the Palaeo-Tethys oceanic crusts (tectonically) which itself was tectonically overlaid by the continental assemblages of the eastern Pontides. On the other hand, Üşümezsoy (1987) stated that Palaeo-Tethys was located at the south of the western Pontides and was closed by the continental collision which took place between the western Pontides and the Menderes Massif during the Late Triassic time. The geological and palaeomagnetic data taken together support the idea that the oceanic remanents of the Palaeo-Tethys may have been caught up in the collision between the eastern and western Pontides (Fig. 8). After the accretion, the Pontides moved as a single tectonic unit from approximately 10°–15° to 23° latitude during the Upper Jurassic–Eocene time (geological evidence for this is the existence of the Upper Jurassic–Lower Cretaceous limestones overlying the entire Pontides). The driving mechanism for this motion may have been the result of the opening of the Neo-Tethys ocean in the south during Upper Jurassic time (Fig. 8). The Neo-Tethys had started subducting under the Pontides, resulting in island arc volcanism on the products of which this study was carried

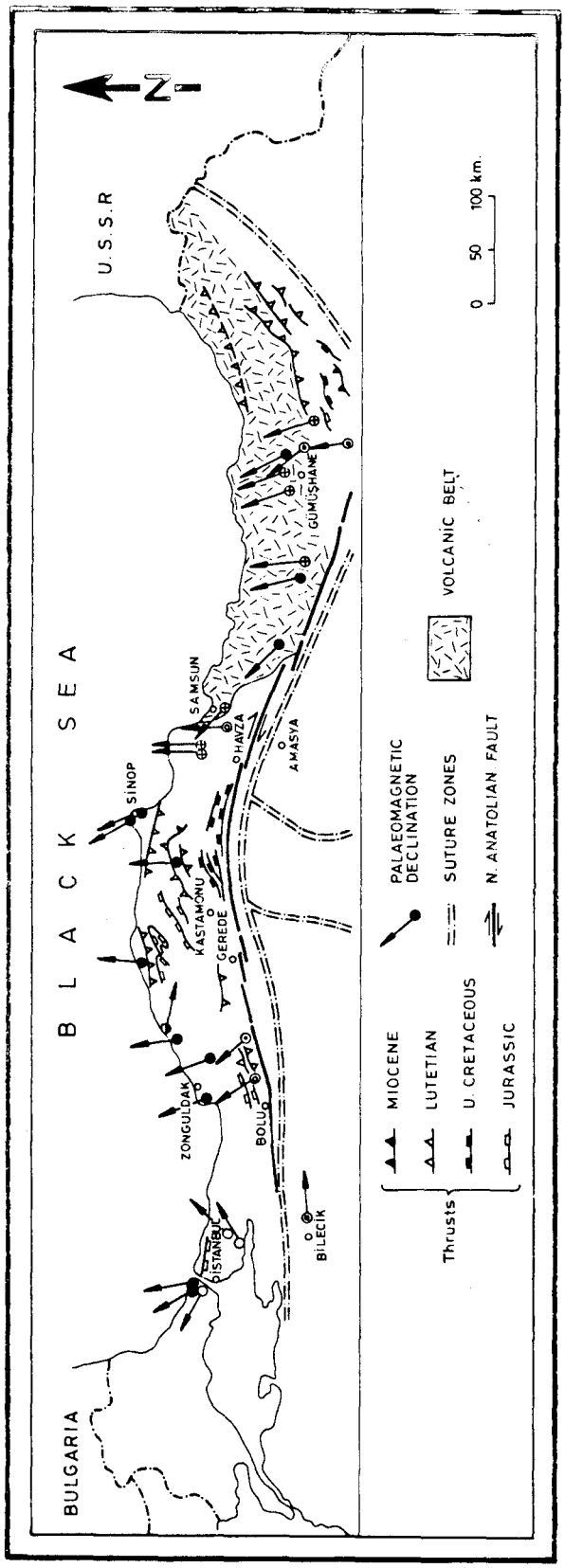


Figure 6. Comparison of the palaeodeclinations with the structural features in the Pontides. The reverse directions are inverted through the origin. The base map is after Ketin (1984); the northern Pontides volcanic belts is taken from Bergougnan & Fourquin (1980). For further symbol explanation, see Fig. 5.

Table 3. Upper Cretaceous and Eocene palaeomagnetic data from previous studies in the Pontides.*

Locality	Age	D (°)	I (°)	References
E Pontides	Upper Cretaceous	153.5	−36.5	Van der Voo (1968)
E Pontides	Upper Cretaceous	140.0	−36.0	Van der Voo (1968)
E Pontides	Upper Cretaceous–Eocene	346.0	40.0	Van der Voo (1968)
E Pontides	Upper Cretaceous	346.8	49.9	Orbay & Bayburdi (1979)
NW Pontides	Upper Cretaceous	351.1	31.4	Lauer (1981)
NW Pontides	Upper Cretaceous	331.6	42.8	Lauer (1981)
E Pontides	Eocene	356.8	48.0	Baydemir (1982)
E Pontides	Eocene	162.0	−42.0	CFR & İTÜ
E Pontides	Eocene	170.0	−45.0	CFR & İTÜ

* Results of Fisher statistics including the present paper for the Upper Cretaceous sites: $N = 13$, $D = 345.6°$, $I = 40.1°$, $k = 34.3$, $\alpha_{95} = 7.2°$; for the Eocene sites: $N = 6$, $D = 351.8°$, $I = 39.6°$, $k = 70.0$, $\alpha_{95} = 8.1°$.

Table 4. Jurassic and Lower Cretaceous data from the Pontides.

Location	Rock type	Age	D (°)	I (°)	References
E Pontides	volcanic	Lower Jurassic	147.0	−4.0	Van der Voo (1968)
E Pontides	limestone	Tithonian	355.0	28.0	CFR & İTÜ
E Pontides	andesite	Aptian	186.3	3.2	This paper
W Pontides	basalt	Kimmeridgian	328.0	9.0	Lauer (1981)
W Pontides	volcanic	Lower Cretaceous	326.0	12.9	Orbay (1978)
SW Pontides	limestone	Upper Jurassic	92.8	53.0	Evans *et al.* (1982)

out. The Eocene palaeolatitude shows no difference from the Upper Cretaceous, suggesting that the northerly drift was terminated by the Upper Cretaceous. In other words, the Pontides reached its present position (~41° latitude) by drifting northwards over 2000 km since the Upper Jurassic.

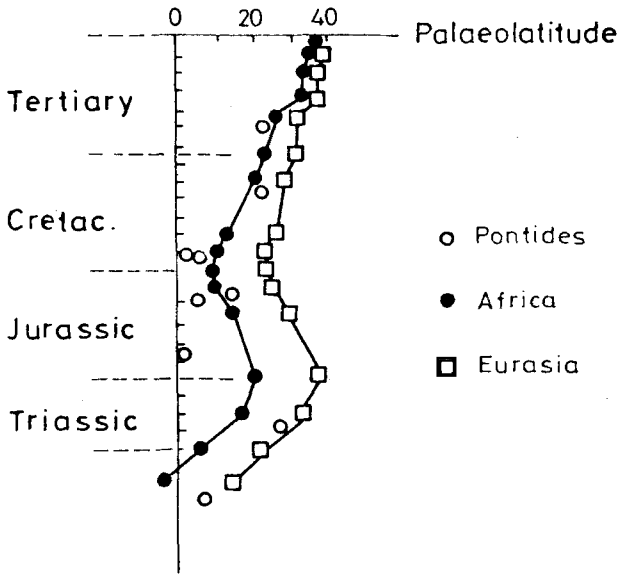


Figure 7. Palaeolatitude curve for the Pontides. The Upper Cretaceous and Eocene palaeolatitudes are estimated from the mean inclination data. For the Jurassic–Lower Cretaceous period, a mean palaeolatitude is not given due to insufficient data; for this reason, a palaeolatitude value corresponding to each site is shown. The Lower Triassic and the Permo-Triassic data are after Sarıbudak *et al.* (1989) and Gregor & Zijderveld (1964), respectively. The Eurasian and African palaeolatitude curves are after Westphal *et al.* (1986).

The available palaeomagnetic data provide evidence as to whether the Pontides belonged to Africa or Eurasia in the past. It is evident from Fig. 7 that palaeolatitudes of the Pontides indicate a position far to the south of Eurasia and close to the African palaeolatitude curve since the Upper Jurassic. Although geological evidence favours a position close to Eurasia (Şengör & Yılmaz 1981; Robertson & Dixon 1984; Görür 1988), palaeomagnetic results suggest that the Pontides remained close to Africa or to the Arabian promontory between Late Jurassic and Late Eocene times.

In this paper my main purpose is to present the new results and to discuss their 'self-evident' tectonic implication and to propose a new hypothesis concerning the region. However, I wish to stress that some of my proposals are still conjectural due to insufficient palaeomagnetic data. In particular, more Lower and Middle Mesozoic data are needed to evaluate the geodynamical evolution of the Pontides more separately.

ACKNOWLEDGMENTS

This paper is a part of PhD study at İTÜ. I am grateful to a very large number of colleagues and friends for valuable discussion and criticism. I am indebted to Professor M. Sanver for his critical review and support during the preparation of the thesis. I particularly thank Dr A. M. C. Şengör for reviewing and improving the text. I thank Drs C. Laj and C. Kissel for having given access to the unpublished data. I am also indebted to the following for the fruitful discussions and their help: Professor Dr N. Canıtez, Dr Ş. Üşümezsoy, E. Ponat and E. İşler. I thank TPAO for providing facilities to carry out field work.

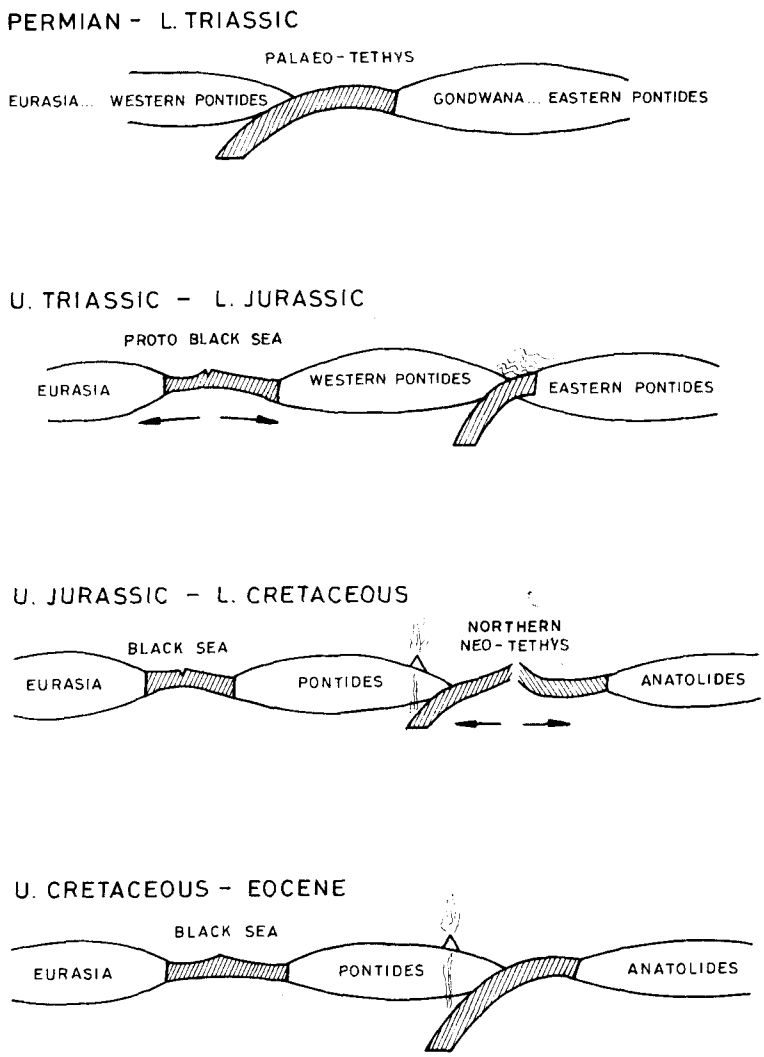


Figure 8. Proposed tectonic evolution of the Pontides between Permian and Eocene time.

REFERENCES

Adamia, S. A., Bergougnan, H., Fourquin, C., Haghipour, A., Lordkipanizde, M. B., Özgül, N., Ricou, L.-E. & Zakairadze, G. S., 1980. The Alpine chaine between the Aegean and the Oman travers, *Colloque C.S, 26° congr. Géol. Int.*, 123–13b, editions du B.R.G.M.

Aydın, M., Serdar, S. & Şahintürk, Ö. 1982. Orta Karadeniz bölgesi jeolojisi ve petrol olanakları, *Türkiye 6. petrol kongresi*, Ankara, 63–71.

Baydemir, N., 1982. Doğu Karadeniz bölgesi Eosen volkaniklerinin palaeomagnetizması, *Doktora tezi*, İstanbul Üniversitesi.

Bazhenov, Z. M. & Burtman, S. V., 1988. Palaeomagnetism of Upper Cretaceous rocks from the Caucasus and its implication for tectonics, in *Tectonic Evolution of the Tethyan Region*, pp. 218–240, ed. Şengör, A. M. C., Kluwer Academic Publishing, Dordrecht, the Netherlands.

Bergougnan, H. & Fourquin, C., 1980. Un ensemble d'élément commus à une marge active alpine des Carpates meriodionales a l'Iran central: le domaine İrano-balkanique, *Bull. Soc. Gèol. France, sèer 7*, **22**, 61–83.

Evans, I., Hall, S. A., Carman, M. F., Şenalp, M. & Çoşkun, S., 1982. A palaeomagnetic study of the Bilecik limestones (Jurassic), North western Anatolia, *Earth planet Sci. Lett.*, **61**, 199–208.

Fisher, B. A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond. Ser. A*, **217**, 295–305.

Görür, N., 1988. Timing of opening of the Black Sea basin, *Tectonophysics*, **147**, 247–262.

Gregor, C. B. & Zijderveld, J. D. A., 1964. The magnetism of some Permian red sandstone from north western Turkey, *Tectonophysics*, **1**, 289–306.

İlhan, E., 1976. *Türkiye Jeolojisi*, O.D.T.Ü. Mühendislik fakültesi yayını.

Ketin, İ. & Gümüş, Ö., 1963. Sinop, Ayancik ve güneyinde III. bölgeye dahil sahaların jeologjisi hakkında rapor: Türkiye Petrolleri A.O., Arama Grubu, rap. no: 213–188.

Ketin, İ., 1966. Anadolunun tektonik birlikleri (Tectonic units of Anatolian Asia minor), M.T.A., Ankara, No: 66, 20–34.

Ketin, İ., 1984. Türkiyenin bindirmeli-naplı yapısında yeni gelişmeler ve bir örnek: Uladağ masifi, *TJK Ketin sempozyumu*, 19–36.

Lauer, J. P., 1981. L'evolution geodynamique de la Turquie et de Chypree Deduite de l'etude palaèomagnetique, *Thèse*, L'institut de Physique du Globe de L'Universite, Louis Pasteour de strasbourg, 299 pp.

Orbay, N., 1978. The palaeomagnetic study of the North Anatolian Fault zone, *İstanbul Üniv. Fen Fak. Mec. seri C.*, **44**, 33–39.

Orbay, N. & Bayburdi, A., 1979. Palaeomagnetism of dykes and tuffs from the Mesudiye region and rotation of Turkey, *Geophys. J. R. astr. Soc.*, **59**, 437–444.

Robertson, A. H. F. & Dixon, J. E., 1985. Aspects of the geological evolution of the eastern Mediterranean, in *Geological Evolution of the Eastern Mediterranean*, Vol. 17, pp. 1–74, eds Dixon, J. E. & Robertson, A. H. F., Blackwell Scientific Publications, Oxford, UK.

Sarıbudak, M., Sanver, M. & Ponat, E., 1989. Location of the western Pontides, NW Turkey, during Triassic time: preliminary palaeomagnetic results, *Geophys. J.*, **96,** 43–50.

Seymen, K., 1975. Kelkit vadisi kesiminde Kuzey Anadolu Fay zonunun tektonik özelliği, *Doktora tezi,* İTÜ., 192 pp.

Şengör, A. M. C., 1979. Mid Mesozoic closure of Permo-Triassic Tethys and its implications, *Nature,* **279,** 590–593.

Şengör, A. M. C., 1984. The Cimmeriden orogenic system and the tectonics of Eurasia, *Geol. Soc. Am., Special Paper 195,* 82.

Şengör, A. M. C., Yılmaz, Y. & Ketin, İ., 1980. Remnants of a pre-late Jurassic ocean in northern Turkey: fragments of Permo-Triassic Palaeo-Tethys, *Bull. geol. Soc. Am.,* **91,** 599–609.

Şengör, A. M. C. & Yılmaz, Y., 1981. Tethyan evolution of Turkey: A plate tectonic approach, *Tectonophysics,* **75,** 181–241.

Şengör, A. M. C., Yılmaz, Y. & Sungurlu, O., 1985. Tectonics of the Mediterranean Cimmerides: Nature and evolution of the western termination of Palaeo-Tethys, in *Geological Evolution of the Eastern Mediterranean,* Vol. 17, pp. 77–112, eds Dixon, J. E. & Robertson, A. H. F., Blackwell Scientific Publications, Oxford, UK.

Şentürk, Ö. & Özçelik, Y., 1983. Zonguldak-Bartın-Amasra-Kurucaşile-Cide dolaylarının jeolojisi ve petrol olanakları, T.P.A.P., Arama Grubu, rap. no: 1816.

Üşümezsoy, Ş., 1987. The NW Anatolian accretionary orogeny; western termination of Palaeo-Tethyan suture belt, *Geol. Bull. Turkey,* **30,** 53–62.

Van der Voo, R., 1968. Palaeomagnetism and the Alpine tectonics of Eurasia, Part 4, Jurassic, Cretaceous and Eocene pole positions from NE Turkey, *Tectonophysics,* **6,** 251–269.

Westphal, M., Mikhail, L. B., Lauer, J. P. Diamar, M. P. & Sibuet, J. U., 1986. Palaeomagnetic implications on the evolution of the Tethys belt from the Atlantic ocean to the Pamirs since the Triassic, *Tectonophysics,* **123,** 37–52.

Yilmaz, Y. & Tüysüz, O., 1984. Kastomonu-Boyabat-Vezirköprü arasındaki bölgenin jeolojisi (Ilgaz-Kargı masiflerinin etüdü), İstanbul Üniversitesi ve M.T.A. ortak raporu.